Chapter 7, Problem 32P

Problem

Evaluate the following integrals:

(a) $\int_1^t u(\lambda)d\lambda$

(b) $\int_0^4 r(t-1)dt$

(c) $\int_1^5 (t-6)^2\delta(t-2)dt$

Step-by-step solution

Step 1 of 2

(a)

Consider the following integral.

$$\int_1^t u(\lambda)d\lambda$$

Simplify the integral.

$$\begin{aligned}\int_1^t u(\lambda)d\lambda &= \int_1^t (1)d\lambda \\ &= \lambda\big|_1^t \\ &= (t-1)\end{aligned}$$

Thus, the simplified expression of the integral is $\boxed{(t-1)}$.

Step 2 of 2

(b)

Consider the following integral.

$$\int_0^4 r(t-1)\,dt$$

Express the signal $r(t-1)$.

$$\begin{aligned} r(t-1) &= (t-1)u(t-1) \\ &= t-1; \qquad t \geq 1 \end{aligned}$$

Simplify the integral.

$$\begin{aligned} \int_0^4 r(t-1)\,dt &= \int_0^4 (t-1)u(t-1)\,dt \\ &= \int_1^4 (t-1)\,dt \\ &= \left(\frac{t^2}{2}-t\right)\Bigg|_1^4 \\ &= \left(\left(\frac{4^2}{2}-4\right)-\left(\frac{1^2}{2}-1\right)\right) \end{aligned}$$

Simplify further.

$$\begin{aligned} \int_0^4 r(t-1)\,dt &= 4-(-0.5) \\ &= 4.5 \end{aligned}$$

Thus, the value obtain from the simplification of the integral is $\boxed{4.5}$.

(c)

Consider the following integral.

$$\int_1^5 (t-6)^2\,\delta(t-2)\,dt$$

Consider the integral property of impulse signal.

$$\int_a^b f(t)\delta(t-t_0) = f(t_0)$$

Simplify the integral.

$$\begin{aligned} \int_1^5 (t-6)^2\,\delta(t-2)\,dt &= \int_1^5 (2-6)^2\,\delta(t-2)\,dt \\ &= (2-6)^2 \int_1^5 \delta(t-2)\,dt \\ &= 16(1) \\ &= 16 \end{aligned}$$

Thus, the value obtain from the simplification of the integral is $\boxed{16}$.